

CS471 – Lesson 6 Solutions Key

Informed Search (Greedy & A*)

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Core Concepts

1. **Heuristic ($h(n)$):**

Answer: A function that estimates how close a state is to a goal. It is designed for a particular search problem.

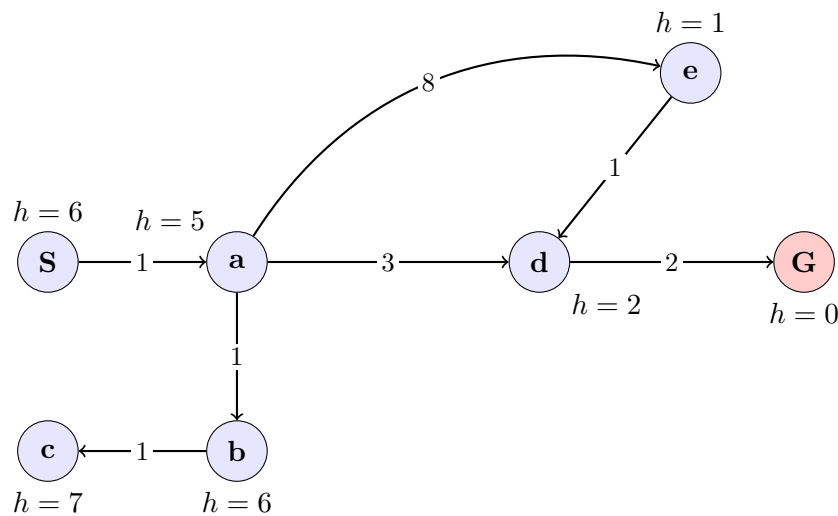
2. **Greedy Search Strategy:**

Answer: A strategy to expand a node that you think is closest to a goal state. It relies solely on the forward cost estimate $h(n)$.

3. **A* Search Strategy:**

Answer: A search that orders the next node by the lowest sum: $f(n) = g(n) + h(n)$. It combines the backward cost $g(n)$ (Uniform Cost Search) with the forward cost $h(n)$ (Greedy).

2. Algorithm Tracing



A. Greedy Search Trace ($f(n) = h(n)$)

Solution

- **Step 1:** Start at S ($h = 6$). Expand to 'a'.
- **Step 2:** From 'a', neighbors are b ($h = 6$), d ($h = 2$), e ($h = 1$). Greedy picks the lowest h , which is 'e'.
- **Step 3:** From 'e', neighbor is 'd' ($h = 2$). Expand to 'd'.
- **Step 4:** From 'd', neighbor is 'G' ($h = 0$). Goal reached!

Final Path: $S \rightarrow a \rightarrow e \rightarrow d \rightarrow G$

Total True Cost: $1 + 8 + 1 + 2 = 12$

Note: This is not optimal!

B. A* Search Trace ($f(n) = g(n) + h(n)$)

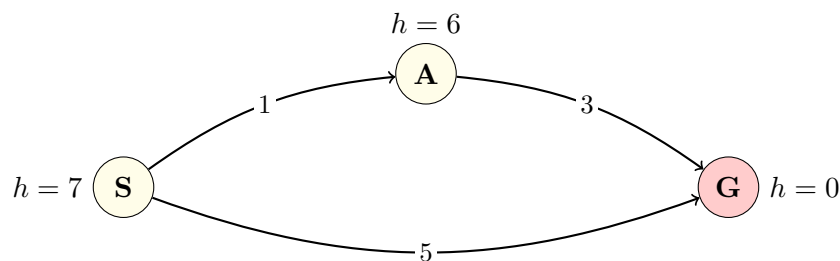
Solution

- **Step 1:** Start at S. $f(S) = 0 + 6 = 6$.
- **Step 2:** Expand 'a'. $f(a) = 1 + 5 = 6$.
- **Step 3:** From 'a', evaluate neighbors:
 - b: $f(b) = (1 + 1) + 6 = 8$.
 - d: $f(d) = (1 + 3) + 2 = 6$.
 - e: $f(e) = (1 + 8) + 1 = 10$.
- **Step 4:** A* picks the lowest $f(n)$, which is 'd' ($f = 6$).
- **Step 5:** Expand 'd' to 'G'. $f(G) = (4 + 2) + 0 = 6$. Goal reached!

Final Path: $S \rightarrow a \rightarrow d \rightarrow G$

Total True Cost: $1 + 3 + 2 = 6$

3. Admissibility and Optimality



Solution

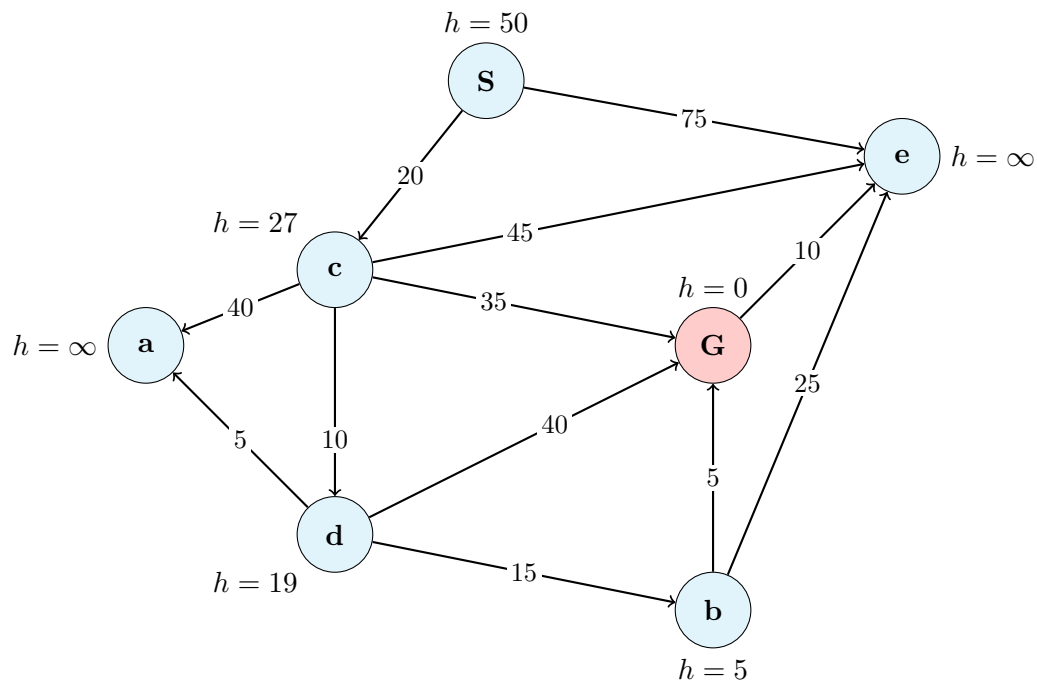
Is the heuristic admissible?

No. An admissible heuristic never overestimates the true cost to reach the goal. We need estimates to be less than actual costs. The true cost from S to G is 4 (via A). However, the heuristic at S is 7 ($7 > 4$). Therefore, the heuristic is NOT admissible.

Why did A* fail?

Because the heuristic is broken, A* calculates $f(S \rightarrow A) = 1 + 6 = 7$. It looks at the direct path $f(S \rightarrow G) = 5 + 0 = 5$. Since 5 is less than 7, A* incorrectly chooses the direct path, resulting in a suboptimal cost of 5 instead of the true optimal cost of 4.

Part 2: Student Independent Practice Solutions



1. Greedy Search Trace ($f(n) = h(n)$)

Solution

- **Step 1:** Start at S ($h = 50$). Expand S.
- **Step 2:** Look at neighbors: c ($h = 27$) and e ($h = \infty$). Greedy picks the lowest h , which is c.
- **Step 3:** Expand c. Look at neighbors: a ($h = \infty$), e ($h = \infty$), G ($h = 0$), d ($h = 19$).
- **Step 4:** Greedy picks the lowest h , which is directly to G. Goal reached!

Final Path: $S \rightarrow c \rightarrow G$

Total True Cost: $20 + 35 = 55$

2. A* Search Trace ($f(n) = g(n) + h(n)$)

Solution

Note: A only tests for the Goal when it DEQUEUEES a node from the Frontier.*

- **Step 1: Pop S.** $f = 0 + 50 = 50$.

- Add c: $g = 20, h = 27 \implies f = 47$

- Add e: $g = 75, h = \infty \implies f = \infty$

Frontier: [c(47), e(∞)]

- **Step 2: Pop c (47).**

- Add d: $g = (20 + 10), h = 19 \implies f = 49$

- Add G: $g = (20 + 35), h = 0 \implies f = 55$ (Wait, do not stop!)

- Add a, e (both $h = \infty, f = \infty$)

Frontier: [d(49), G(55), a(∞), e(∞)]

- **Step 3: Pop d (49).**

- Add b: $g = (30 + 15), h = 5 \implies f = 50$

- Add G: $g = (30 + 40), h = 0 \implies f = 70$

- Add a ($h = \infty$)

Frontier: [b(50), G(55 via c), G(70 via d), ...]

- **Step 4: Pop b (50).**

- Add G: $g = (45 + 5), h = 0 \implies f = 50$

- Add e ($h = \infty$)

Frontier: [G(50 via b), G(55 via c), G(70 via d), ...]

- **Step 5: Pop G (50).** Goal reached!

Final Path: $S \rightarrow c \rightarrow d \rightarrow b \rightarrow G$

Total True Cost: $20 + 10 + 15 + 5 = 50$

Note: A finds the optimal path (50), while Greedy got trapped into a suboptimal path (55) because 'c' visually looked closest to G.*